

MOHD SAIFUDDIN

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SUMMARY

I possess strong technical skills in the AI/ML domain and am keen to explore opportunities where I can leverage my expertise. I have built multiple projects, including fine-tuning LLaMA 3.2 using QLoRA, deploying a real-time emotion detection pipeline on CCTV feeds, and developing a RAG system using FAISS and Flan-T5.

EDUCATION

Rajiv Gandhi Institute of Petroleum Technology
Institute of National Importance
B.Tech in Electronics Engineering; CPI: 7.3/10.0

Jais, India
May 2026

TECHNICAL SKILLS

Languages: Python, C++, SQL

ML & DL: PyTorch, TensorFlow, Hugging Face Transformers, Sentence Transformers, QLoRA, PEFT, DeepFace, MTCNN

NLP & LLMs: LLM Fine-Tuning (Llama, DeBERTa), RAG, FAISS, Flan-T5, Unsloth

Tools & Infrastructure: FastAPI, AWS, Docker, Git, Selenium, NumPy, Pandas, OpenCV

EXPERIENCE

Asakta Electronics and Communication
Data Engineering Intern

Hyderabad, India
May 2025 – July 2025

- Worked at a smart governance company specializing in energy-sector IoT and smart metering solutions.
- Engineered real-time ingestion APIs (FastAPI, AWS) to process telemetry from 150K+ IoT devices, leveraging async pipelines and Pydantic validation to ensure 100% data integrity.
- Reduced system latency and improved throughput by deploying high-concurrency async services for scalable cloud streaming workloads.
- **Stack:** Python, SQL, FastAPI, Kafka, AWS, Parallel Programming.

PROJECTS

Real-Time Emotion Intelligence Pipeline 🌀 | Python, MTCNN, DeepFace, TensorFlow, OpenCV Mar 2026

- Built a production-grade **facial emotion analysis** system for CCTV footage using **MTCNN** for face detection and **DeepFace (FaceNet 128d)** for identity tracking via cosine similarity, achieving strict **offline privacy compliance**.
- Designed memory-safe video processing with Python generators, **4x spatial downscaling**, and **micro-expression detection** at **8 FPS** temporal resolution, outputting emotion timelines and stress indexes to a self-contained HTML dashboard.

LLM Human Preference Prediction 🌀 | PyTorch, Llama 3.2, DeBERTa, QLoRA, Unsloth Jan 2026 – Feb 2026

- Fine-tuned a hybrid ensemble (**Llama 3.2 3B + DeBERTa-v3**) on **57K+ chatbot interactions** for the LMSYS Kaggle competition, replacing Llama's causal head with a **custom 3-class PyTorch classifier** to predict human preferences.
- Reduced GPU memory from 48GB to under 4GB using **4-bit QLoRA** via Unsloth, trained on **RunPod A6000**, and built a fully **offline inference pipeline** achieving **1.10 Log Loss**.

RAG Question-Answering System 🌀 | Python, FAISS, Sentence Transformers, Flan-T5 Feb 2025 – Mar 2025

- Scraped **100 government schemes** from myscheme.gov.in using **Selenium**, chunked and embedded text with **all-MiniLM-L6-v2** (384d), and indexed into a **FAISS** vector store for sub-second retrieval.
- Built an interactive **RAG pipeline** connecting retrieved context to **Flan-T5-Large** for answer generation, delivering accurate natural language responses with **30% retrieval accuracy improvement** over naive search.

SAR-to-Optical Image Translation (GAN) 🌀 | PyTorch, S-CycleGAN, Computer Vision Sept 2023 – Dec 2023

- Trained a **Supervised CycleGAN** on **20,000+ paired SAR-optical images** to translate satellite radar data into photorealistic optical formats, validated with **PSNR** and **SSIM** metrics.

LEADERSHIP & ACHIEVEMENTS

IEEE RGIPT: Technical Branch Head (Student Chapter) | **Hackathon Winner:** IEEE Kode Kurrent 1.0 |

Mentoring: Gyan Arpan (student mentor).